

Homework 3

Problem 1: Sequence Comparison using RNNs (20 pts)

Consider a recurrent neural network with scalar hidden states and integer inputs. All biases are zero. The hidden state evolves according to the following recurrence:

$$h_t = \begin{cases} h_{t-1} + x_t & t \leq T/2 \\ h_{t-1} - x_t & t > T/2 \end{cases} \quad \text{with } h_0 = 0$$

The output unit performs binary classification at the final time step:

$$y = \sigma(h_T)$$

Assume that the input sequence is of even length.

1. Write h_T explicitly in terms of the input sequence $\{x_1, x_2, \dots, x_T\}$.
2. What function of the input sequence is computed by the hidden state at the final time step? Be precise.
3. Suppose $x_t \in \{0, 1\}$. Under what condition will the output y be close to 1? Under what condition will it be close to 0?

Problem 2: Attention! My code takes too long (20 pts)

In class, we showed that computing a regular self-attention layer takes $O(T^2)$ running time for an input with T tokens. Propose two different ways to reduce this running time to $O(T)$, and comment on their possible pros vs. cons.

Problem 3: Minimax Optimization (20 pts)

In this problem we will see how training a GAN is fundamentally different from regular training. Consider a simple problem where we are trying to minimax a function of two scalars. The function is given by:

$$\min_x \max_y f(x, y) = 4(x - 1)^2 - 4(y + 2)^2$$

You can try graphing this function to get a sense of what it looks like in Python, but you do not need to submit the code.

1. Determine the saddle point of the function $f(x, y)$. A saddle point is a point for which f attains a local minimum along one direction and a local maximum along another.

2. Write down the gradient descent/ascent equations for solving this problem starting at some arbitrary point (x_0, y_0) .
3. Determine the range of allowable step sizes to ensure that gradient descent/ascent converges to the saddle point.
4. What if you just did regular gradient descent over both x and y ? Would you still converge to the saddle point? Why or why not? Comment on the dynamics of the updates and whether there are special cases where you might converge to the saddle point.
5. Why do we do a minmax instead of a maxmin optimization?

Problem 4: GAN vs. Conditional GAN (20 pts)

In this problem we will modify the demo from the Friday recitation class to train both a GAN and a Conditional GAN on FashionMNIST. The demo from class is available at the Demo GitHub repo. You may reuse code from the demo; clearly mark your changes.

(a) Train a standard GAN on FashionMNIST (8 pts)

Train a GAN using the same architecture as in the demo, but without conditioning.

(b) Train a Conditional GAN on FashionMNIST (6 pts)

Train a Conditional GAN where the generator and discriminator are conditioned on class labels.

(c) Research some of the measures of GAN quality and report them for your trained GAN, if applicable.

(d) Compare GAN vs. Conditional GAN (6 pts)

- How conditioning affects the quality and diversity of generated samples.
- Whether the Conditional GAN can generate specific classes reliably.
- Differences in training stability between the two models.

Problem 5: A Tiny Diffusion Model on FashionMNIST (20 pts)

In this problem you will extend the Diffusion demo notebook in the course GitHub repo and compare it to your GAN on FashionMNIST. You may reuse code from the demo; clearly mark your changes.

(a) FashionMNIST instead of MNIST (4 pts)

Train the diffusion model on FashionMNIST:

- Replace MNIST with FashionMNIST and keep the expected normalization.
- Train for a few epochs / reasonable number of iterations.

Submit:

1. 2–4 sentences describing your code changes.
2. One 4×4 grid of generated FashionMNIST samples.
3. 1–2 sentences commenting on sample quality.

(b) Forward diffusion visualization (6 pts)

Visualize how data evolves under the forward diffusion process.

1. Show images at multiple timesteps (e.g., $t = 0, 100, 300, 500$).

2. Describe how increasing noise affects different components of the image (e.g., edges, shapes, and pixel intensity variations), and identify which components are most sensitive to noise.

(c) Noise ablation on diffusion (4 pts)

Compare the default linear β_t to a nonlinear schedule (e.g., quadratic or cosine-like) with fixed T .

1. Visualize how data evolves under the forward diffusion process.
 - (a) A plot of training loss vs. iteration/epoch for the two settings.
 - (b) Two 4×4 sample grids (one per setting).
 - (c) 3–5 sentences interpreting effects on training and sample quality; relate briefly to the lecture intuition about noise level.

(d) GAN vs. Diffusion on FashionMNIST (6 pts)

Compare your FashionMNIST diffusion model to your FashionMNIST GAN from Problem 4. Submit:

1. A short writeup (roughly half a page) on:
 - Training dynamics (stability, tuning effort).
 - Sample quality (sharpness vs. diversity) with one grid of GAN samples and one grid of diffusion samples.
 - A brief link to theory: GANs as minimax/adversarial training vs. diffusion as noise-prediction/denoising with a fixed loss.